

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 01



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Topics to be Covered



Topic

Sets and representation of sets



Topic

Different types of sets



Topic

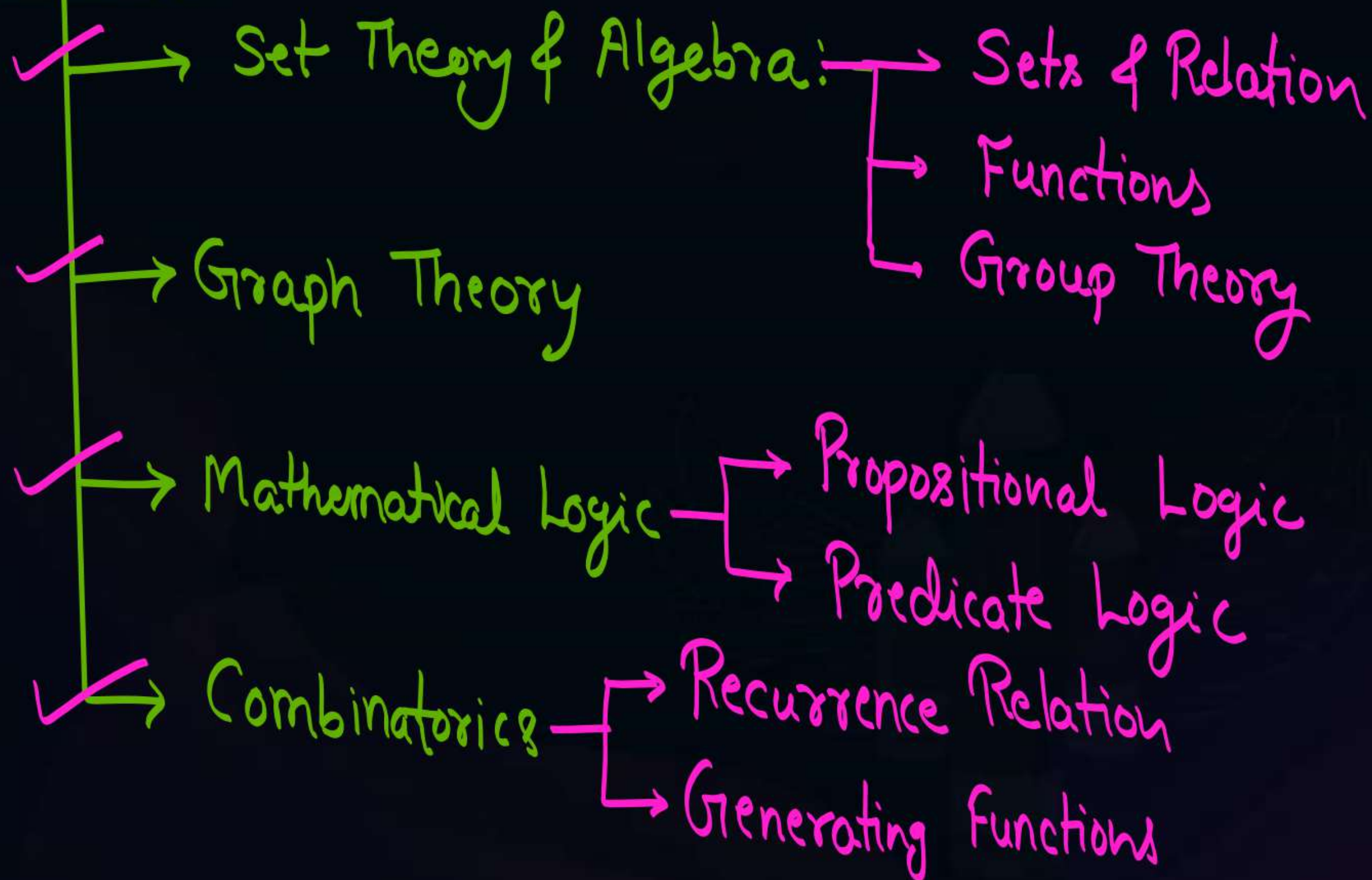
Number of subsets of a given set





Topic : Syllabus

Weightage: Gate-2026 → Approximately
(7-9) Marks





Topic : Set



- A well-defined unordered collection of distinct elements is called as set. ✓

$$\text{eg. } A = \{1, 2, 3, a, \text{Jan}\} = \{\text{Jan}, 1, 2, a, 3\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$C = \{a, e, i, o, u\}$$



Topic : Representation of sets

There are three different ways in which set may be represented.

- ✓ 1. Roaster form or Tabular form:
- ✓ 2. Set-builder form:
- ✓ 3. Statement form:



Topic : Roaster form or Tabular form

We will list all the elements of the set within the curly braces.

eg: $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, \text{Jan, feb}\}$

$X = \{1, 2, 3, 4, 5, \dots\}$

We may not represent the elements of set B using "set builder" form



Topic : Set-builder form

- In this notation we will define the property that must be satisfied by the elements of that set.

$$C = \{x : x \in \mathbb{N} \text{ and } x \leq 5\}$$

Such that Natural number

$$D = \{y \mid y \in \mathbb{R} \text{ and } 0 \leq y \leq 1\}$$

Real number

{ We may not represent the elements of set D using Tabular form }



Topic : Statement form

$A =$ Set of all natural numbers less than
or equal to 5.

$D =$ Set of real numbers between 0 & 1



Topic : Cardinality of a set

{ Generally "Cardinality" is }
{ defined only for finite sets } PW

Cardinality of set 'A' is denoted by ' $|A|$ ',

and it represent the number of elements
in set A.

eg. $A = \{ 1, 2, 3, \text{Jan, Feb} \}$

$$|A| = \text{No. of elements in set A.}$$
$$= 5$$



Topic : Types of sets

Empty Set: A set with no elements in it is called empty set, it is denoted by $\emptyset$ or $\{ \}$

$$\emptyset = \{ \}$$

$$|\emptyset| = |\{ \}| = 0$$

$$|\{ \underline{1}, \underline{\{2, 3\}} \}| = 2$$



Topic : Types of sets

Singleton Set: A set with exactly one element in it is called a singleton set.

eg. $\{1\}$, $\{a\}$, $\{\underbrace{\{a, b, c\}}_{\text{whole set is treated as a single element}}\}$

$$|\{\emptyset\}| = |\{\underbrace{\{\}}\}| = 1$$

$$\emptyset = \{\}$$

$\{\}$
Empty set

$$\Downarrow$$

$$|\{\}| = 0$$

$$\{\underbrace{\{\}}\} \Rightarrow |\{\{\}\}| = 1$$

A set containing
one element, and
that element is
an empty set

$$\{\} \neq \{\{\}\}$$



Topic : Types of sets

Finite Set:

A set that contains finite number of elements in it is called a finite set.

g. $\{\}$, $\{1\}$, $\{1, a\}$, $\{a, 1, 3\}$

↳ Empty set is always a finite set.



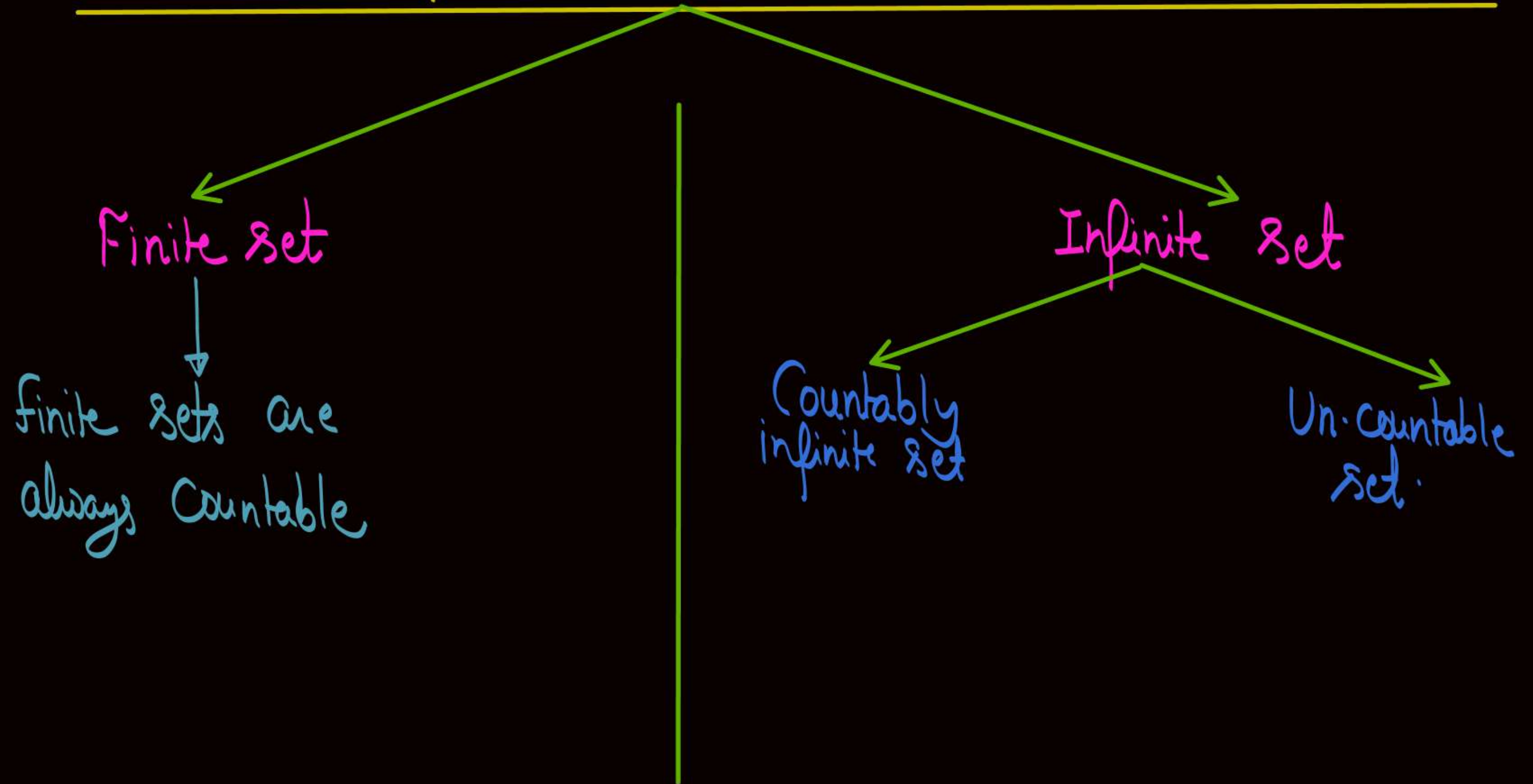
Topic : Types of sets

Infinite Set: A set that contains infinite number of elements is called an infinite set.

eg. $A = \text{Set of all natural numbers}$
 $A = \{1, 2, 3, \dots, 2000, \dots\}$
 $B = \{y \mid y \in \mathbb{R} \text{ and } 0 \leq y \leq 1\}$

Both are infinite sets

Another classification w.r.t. number of elements in the set





Topic : Types of sets

- ★ Equal Sets: Two sets A and B are said to be equal if and only if they have exactly same elements.
 {i.e. Every element of set A must be present in set B. }
 {And Every element of set B must be present in set A. }
- eg. $A = \{1, 2, 3, a, b\}$, $B = \{a, 1, 2, b, 3\}$
 here $A = B$

eg. $A = \{1, 2, 3, 4\} \Rightarrow A \neq B$
 $B = \{1, 2, 3, a\}$

eg. $A = \{1, 2, 3, 4, 5\} \Rightarrow A \neq B$
 $B = \{1, 2, 3, 5\}$



Topic : Types of sets

Equivalent Sets: Two finite sets A and B are said to be Equivalent sets if and only if $|A| = |B|$
i.e. Both have same Cardinalities.

eg. $A = \{1, 2, 3, a, b\}$, $B = \{a, 1, 2, b, 3\}$

$|A| = 5$ & $|B| = 5 \Rightarrow \therefore A$ is equivalent to B .

Here $A = B$ as well

eg. $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, a\}$

Here $A \neq B$, but $|A| = |B| = 4 \therefore A$ is equivalent to B .

eg. $A = \{1, 2, 3, 4, a\}$, $B = \{1, 2, a, b\}$

Here $A \neq B$, and $|A| = 5$ & $|B| = 4$

ie, $|A| \neq |B| \therefore A$ is not equivalent to B .

Note:-

① If two sets are equal, then they are equivalent as well.

② If two sets are not equal, then they may or may not be equivalent.

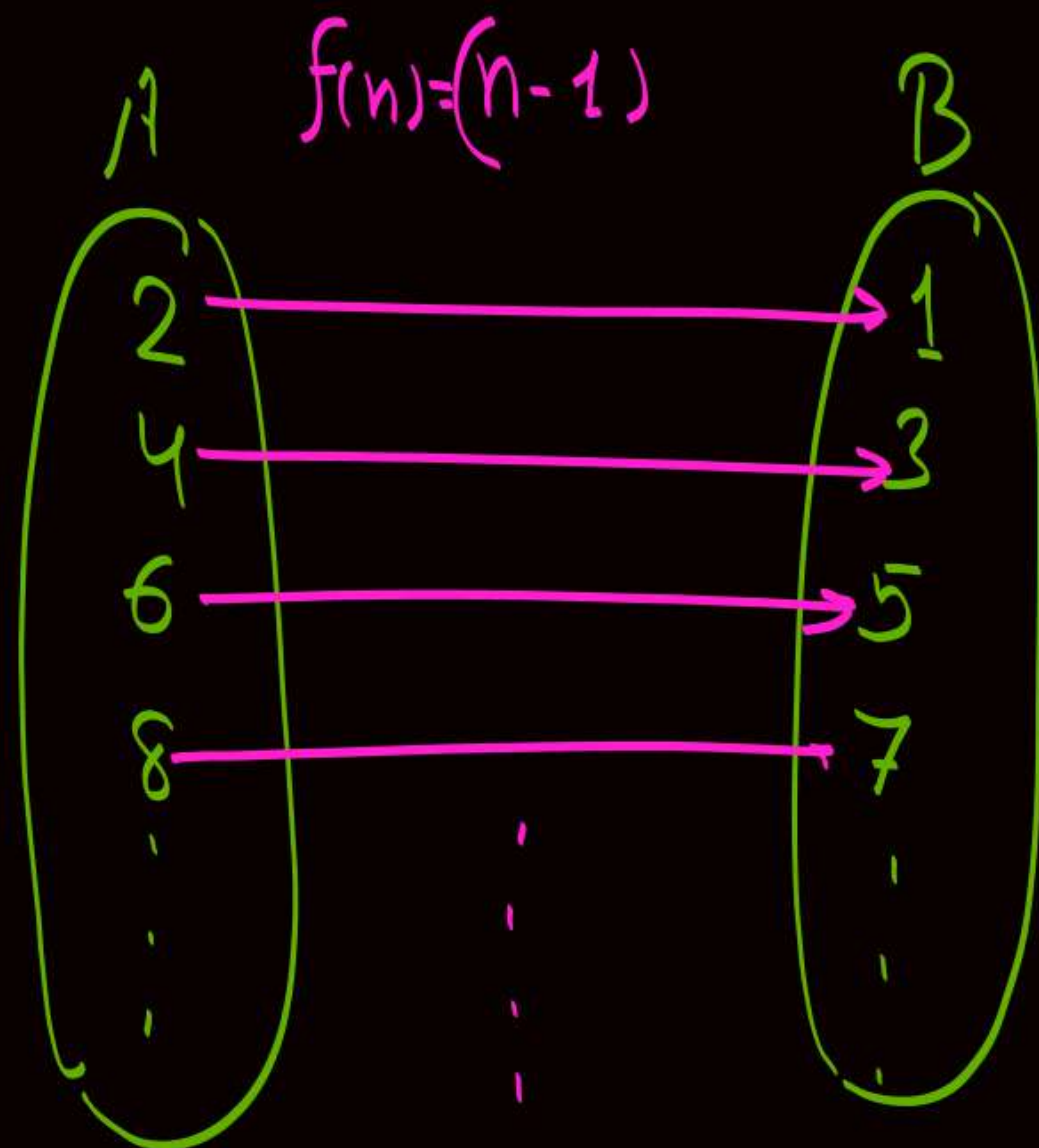
③ If two sets are not equivalent, then they can never be equal.

* Equivalent sets :- Two sets A and B are said to be equivalent sets if and only if there exists a bijection (one-one and onto function) between them

eg. A = Set of all non-zero, non-negative even integers =

B = Set of all non-zero, non-negative 'Odd' integers.

$$f: A \rightarrow B, f(n) = (n-1)$$



$$f^{-1} : B \rightarrow A$$
$$f^{-1}(n) = (n+1)$$

it is bijective
 $\therefore A \& B$ are equivalent.



Topic : Types of sets

Universal Set: It is the set of all elements associated with the problem under consideration.

generally universal set is denoted by ' U '.



Topic : Types of sets

Subset & Superset:

" $\subseteq$ "

" $\supseteq$ "

Let A and B are any two sets,
if every element of set 'A' is included
in set 'B' as well, then we say that
A is subset of B {i.e. $A \subseteq B$ }, or B
is superset of A, {i.e. $B \supseteq A$ }

A is a subset of B
if and only if
B is a superset of A

Note:-

① If $A \subseteq B$ and $B \subseteq A$,
then $A = B$

Note:

In subset ($\subseteq$) Equality Condition is also included

∴ Every set is a subset of itself.

i.e., $A \subseteq A$ is always true for any set A

Note:-

$\emptyset$ is a subset of Every set,

i.e., $\emptyset \subseteq A$ is always true for any set A .



Topic : Types of sets

Proper Subset:



Equality Condⁿ
is not included.

For a given set A , any subset of set A except set A itself are called proper subsets of set A .

→ A is proper subset of B . {i.e. $A \subset B$ }
if and only if

B is proper superset of set A . {i.e. $B \supset A$ }

① if $A \subseteq B$ and $B \not\subseteq A$, then $A \subset B$

② if $A \not\subseteq B$ and $B \subseteq A$, then $B \subset A$

eg. Let $A = \{1, 2, 3\}$

Subsets of Set 'A' are $= \{ \}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

Proper subsets of set A are $= \{ \}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$

$\{ \text{it is included in subsets} \}$
 $\{ \text{But, Not included in proper subsets} \}$



Topic : Number of subsets of a set 'A' of cardinality 'n'

* Let 'A' is a finite set of size = n , { i.e. Cardinality of $A = n$ }
i.e., $|A| = n$

How many subsets of set 'A' are possible = ?

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

eg. Let $A = \{1, 2, 3\}$, $|A| = 3$

↳ Every subset of set A will include '0' or more of those three elements but no extra element.

Subsets of A
of size = 0

↓
{ }

Number of subsets
of size '0' = ${}^3C_0 = 1$

Subsets of A
of size = 1

{1}, {2}, {3}

Number of subsets
of size '1' = ${}^3C_1 = 3$

Subsets of A
of size = 2

{1, 2}, {1, 3}, {2, 3}

Number of subsets
of size '2' = ${}^3C_2 = 3$

Subsets of A
of size = 3

↓
{1, 2, 3}

Number of subsets
of size '3' = ${}^3C_3 = 1$

∴ Total no. of subsets of set A = ${}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3 = 1 + 3 + 3 + 1 = 8$

Note :-

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

Note :

$$(1+x)^n = {}^nC_0(x)^0 + {}^nC_1(x)^1 + {}^nC_2(x)^2 + \dots + {}^nC_n(x)^n$$

Put $x=1$,

$$(1+1)^n = {}^nC_0(1)^0 + {}^nC_1(1)^1 + {}^nC_2(1)^2 + \dots + {}^nC_n(1)^n$$

$$(2)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$$



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How many subsets of set 'A' are possible = ?

$$\begin{aligned} \text{Total number of subsets of set A} &= \text{Subsets of size '0'} + \text{Subsets of size '1'} + \text{Subsets of size '2'} + \dots + \text{Subsets of size 'k'} + \dots + \text{Subsets of size 'n'} \\ &= {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_k + \dots + {}^nC_n = 2^n \end{aligned}$$



Topic : Number of subsets of a set 'A' of cardinality 'n'

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i.e., $|A| = n$

How many subsets of set 'A' are possible = ?

Total number
of subsets
of set A

$$= {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_k + \dots + {}^nC_n = 2^n$$

Number of subsets of
set 'A' s.t. size of
each subset is
at most 'k'

Number of subsets
of size exactly = k

Number of subsets
of set A such that
size of each subset
is at least 'k'.



2 mins Summary

time/vishal Rawtiyapu



✓
Topic

Sets and representation of sets

✓
Topic

Different types of sets

✓
Topic

Number of subsets of a given set

THANK - YOU